Useful distributions

- Normal(mean $= \mu$, variance $= \sigma^2$). The Normal distribution with mean μ and variance σ^2 has the probability density function

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  and has the expectation $\mathrm{E}(X) = \mu$.

- Exponential(rate $= \alpha$). The exponential distribution with rate $\alpha > 0$ has the probability density function

$$f(x; \alpha) = \alpha e^{-\alpha x}$$

  for $x \geq 0$, and has the expectation $\mathrm{E}(X) = \frac{1}{\alpha}$.

- Gamma(shape $= a$, rate $= b$). The Gamma distribution with shape $a > 0$ and rate $b > 0$ has the probability density function

$$f(x; a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$$

  for $x > 0$, and has the expectation $\mathrm{E}(X) = \frac{a}{b}$.

- Inverse-Gamma(shape $= a$, scale $= b$). The Inverse-Gamma distribution with shape $a > 0$ and scale $b > 0$ has the probability density function

$$f(x; a, b) = \frac{b^a}{\Gamma(a)} x^{-a-1} e^{-\frac{b}{x}}$$

  for $x > 0$, and has the expectation $\mathrm{E}(X) = \frac{b}{a-1}$ for $a > 1$.

- Poisson(rate $= \lambda$). The Poisson distribution with rate $\lambda > 0$ has the probability mass function

$$f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

  for $x = 0, 1, \ldots$, and has the expectation $\mathrm{E}(X) = \lambda$.

- Categorical(π_1, π_2). The categorical distribution with event probabilities π_1 and π_2, where $\pi_1 > 0$, $\pi_2 > 0$, and $\pi_1 + \pi_2 = 1$, has the probability mass function

$$f(x; \pi_1, \pi_2) = \pi_x$$

  for $x = 1, 2$, and has the expectation $\mathrm{E}(X) = 2 - \pi_1$.

- Categorical(π_1, π_2, π_3). The categorical distribution with event probabilities π_1, π_2, and π_3, where $\pi_1 > 0$, $\pi_2 > 0$, $\pi_3 > 0$, and $\pi_1 + \pi_2 + \pi_3 = 1$, has the probability mass function

$$f(x; \pi_1, \pi_2, \pi_3) = \pi_x$$

  for $x = 1, 2, 3$.